

DIABETES PREDICTION

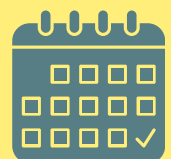
CAPSTONE PROJECT
BUSINESS PRESENTATION



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POST GRADUATE DIPLOMA IN ARTIFICIAL
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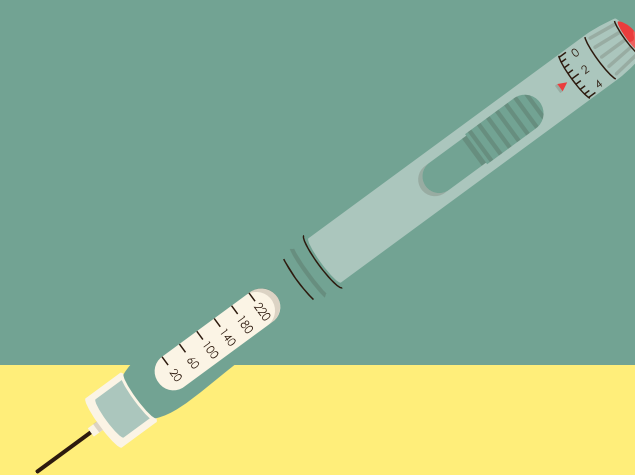


JULY 17, 2026



What are we doing?

Predict diabetes using health indicators



Why are we doing this?

TO SAVE LIVES.

830,000,000 people have diabetes (2022)

Rapidly rising in low and middle income countries

2,000,000 deaths in 2021 due to diabetes & kidney disease. 11% caused by high blood glucose.

*WHO Statistics**

PREVENTION IS BETTER THAN CURE.

\$640B - annual cost of diabetes in the USA
\$5,876/person - medicare 65yo and above

<https://www.cdc.gov/nccdphp/priorities/diabetes-interventions.html>

Philippines

Outpatient annual cost **USD807 to USD2,507**

<https://pmc.ncbi.nlm.nih.gov/articles/PMC8506878/>

Jimeno, C., Sy, R. A., De La Pena, P., Cipriano, C., Tan, R., Panelo, A., & Ng, J. Y. S. (2021). Direct medical costs of type 2 diabetes mellitus in the Philippines: findings from two hospital databases and physician surveys. *BMJ Open*, 11(10), e049737. doi.org

HOW?

Pima Indians Diabetes Dataset

768 FEMALE PATIENTS
AT LEAST 21 YEARS OLD
8 FEATURES PLUS OUTCOME

kaggle™

Indicators

PREGNANCIES
GLUCOSE
BLOOD PRESSURE
SKIN THICKNESS
INSULIN
BMI
DIABETES PEDIGREE FUNCTION
AGE

EDA - Exploratory Data
Analysis Notebook

MODEL IMPLEMENTATION

Bias and Fairness Review

RESULTS

MODEL ACCURACY

94.7%

Our AI model correctly identifies diabetic patients 94.7% of the time when given a choice between a diabetic and a non-diabetic patient.

To put that in context – a coin flip would score 50%. A good clinical screening tool typically scores above 80%. Ours scores 94.7%

RECALL

85/100

For every 100 patients who actually have diabetes, our model correctly flags 85 of them for follow-up care – before they even show symptoms.

Compare that to current standard screening, which catches roughly 52 in 100. That's a 63% improvement in early detection.

PRECISION

80.7%

When the model flags a patient as high risk, it is correct about 80% of the time. Roughly 1 in 5 flagged patients will turn out not to have diabetes – they'll simply need a follow-up consultation to confirm.

This is an acceptable false alarm rate in a clinical setting – far better than over-screening everyone.



The Model Explains Itself – Clinicians Can Trust It

SHAP analysis confirmed Glucose (#2) and Insulin (#1) as top drivers, BMI and Age as mid-tier contributors.

When the model flags a patient, the clinician sees exactly why – *This patient was flagged primarily because their glucose level is in the diabetic range and their BMI is above 33.*

Doctors aren't asked to trust a mysterious algorithm. They're given clear, clinically familiar reasons they can act on immediately.

Clinical adoption fails when doctors don't trust the tool. Explainability is also increasingly required by healthcare regulators. This feature directly reduces adoption risk and accelerates deployment.



Passed the Fairness Audit

The model was audited for fairness across different patient groups – checking whether it performs equally well for older and younger patients, across different body types, and across pregnancy histories.

The model passed on two out of three dimensions. The one area to watch is younger patients aged 21 to 30 – the model is slightly less accurate for this group because there are fewer young diabetic patients in the training data. We've flagged this group for mandatory clinician review and are prioritising collecting more data from younger patients going forward.

Healthcare AI bias is a serious legal and reputational risk. Since we have proactively audited – and have a clear plan for the weak area – demonstrates responsibility and significantly reduces liability exposure.

TARGET BUSINESS AUDIENCE



Hospitals & Health Systems

- Reduce readmissions and late-stage treatment burden



Public Health Ministries

- Population-scale screening at low cost per patient



Health Insurers

- Reduce high-cost claims from preventable complications



Corporate Wellness Programmes

- Protect workforce productivity and reduce insurance premiums



Medical Device Companies

- Larger early-diagnosis market for glucose monitors and devices



Digital Health & Telehealth Platforms

- Embed as a pre-consultation risk screening feature



Common concern: ***"Will earlier detection reduce drug sales?"***

THE REALITY: Currently undiagnosed: 50% of diabetic patients globally

→ These patients buy ZERO medication today

→ Early detection creates new prescription patients

Fastest growing drug categories are EARLY-STAGE:

→ GLP-1 agonists (Ozempic, Wegovy): \$50B+ market by 2030

→ SGLT2 inhibitors: prescribed at pre-diabetes stage

→ Metformin: first-line early treatment

Pharma companies already investing in early detection:

→ Novo Nordisk, Eli Lilly, Roche all have digital health divisions

→ Earlier diagnosis = earlier brand loyalty = longer treatment duration

CONCLUSION: Early detection expands the treated population.

The pharmaceutical opportunity GROWS, not shrinks.



SHORT TERM BUSINESS PLANS

1. Pilot Programme in 2-3 Clinics

Deploy the model as a decision support tool at triage in selected clinics. Measure detection rate improvement, clinician adoption rate, and false alarm burden. This generates real-world evidence to support wider rollout.

KPI: Detect 25% more diabetic patients vs pre-pilot baseline within 6 months.

2. Electronic Health Record Integration

Work with EHR vendors to embed the model as a risk flag that appears automatically during patient intake – requiring no extra clinician effort.

KPI: Zero additional clinical workflow steps required to generate a risk score.

3. Clinical Champion Programme

Identify 2-3 senior clinicians who believe in the tool and can advocate internally. Peer trust is the fastest path to clinical adoption – doctors trust other doctors more than they trust data scientists.

KPI: At least 80% of flagged patients reviewed and acted upon within the pilot period.

4. Regulatory Pre-Submission

Begin engagement with the relevant medical device regulatory body. AI clinical decision support tools increasingly require regulatory clearance – starting early avoids delays at scale.

KPI: Regulatory pathway identified and pre-submission meeting scheduled.

5. Model Retraining on Local Data

The Pima dataset is a proof of concept. The model must be retrained on the local patient population before real deployment. Collect consented patient data with outcome labels and retrain within the first 6 months.

KPI: Retrained model validated on local cohort with AUC-ROC ≥ 0.90 .

LONG TERM BUSINESS PLANS

1. Population-Scale Deployment

Expand from pilot clinics to an entire health system, region, or national screening programme. At scale the ROI becomes transformational – screening millions of patients annually with minimal incremental cost per screen.

KPI: 500,000+ patients screened annually by Year 3.

2. Continuous Learning Model

Implement an MLOps pipeline where the model retrains automatically on new patient data – improving over time as it sees more cases. Include drift monitoring to detect when model performance degrades.

KPI: Quarterly automated retraining cycle with performance dashboard.

3. Expand to Related Conditions

Diabetes is one of a cluster of metabolic conditions – pre-diabetes, hypertension, cardiovascular disease, and obesity are closely linked. The same modelling framework can be extended to predict these conditions from similar clinical inputs.

KPI: 3 additional condition models deployed by Year 5.

4. Integration with Wearables & Remote Monitoring

Connect the model to continuous glucose monitors, smartwatches (heart rate, activity), and remote blood pressure monitors. This enables real-time risk scoring rather than point-in-time screening – flagging patients as their risk changes between appointments.

KPI: Live risk score updated daily per patient from wearable data feeds.

5. Preventive Care Pathway Automation

Move beyond flagging patients to automating the preventive care response – automatically booking follow-up appointments, sending dietary guidance, or triggering a nurse outreach call when a patient is flagged. Closes the loop between prediction and action.

KPI: Time from patient flag to preventive care intervention reduced from weeks to 48 hours.

6. Licensing the Model as a SaaS Product

Package the trained model and deployment infrastructure as a Software-as-a-Service product sold to other healthcare providers – creating a recurring revenue stream beyond the initial deployment.

KPI: 10+ healthcare organisations licensed by Year 4.

THANK YOU!

github: superk314

